S C - 3 0 0

C O M M A N D

R E F E R E N C E

M A N U A L

P R E L I M I N A R Y

D R A F T

Do Not Duplicate or Redistribute
Without Written Permission
By the GE Company

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

TABLE OF CONTENTS

1.0 INTRODUCTION **3**
2.0 DISPLAY COMMAND GROUP **4**
DIS - Display a new screen 5
PRE - Display previous screen 6
PGU - Advance to next display 7
PGD - Back to previous display 7
ESC - Simulate ESCAPE Key pressed 8
SHO - Show background format 9
ADD - Create dynamic data definitions 10
DEL - Delete a display file from disk 12
REM - Delete a format file from disk 13
KIL - Remove a display from memory 14
FRE - Delete a format from memory 15
CLE - Clear the current display 16
DRA - Enter screen drawing utility 17
CRE - Create a background format 18
REF - Refreshes the current display 19
PAR - Dynamic data parameter commands 20
PARADD - Add a new display parm. 20
PARDEL - Delete a display parm. 20
PARDOC - Create DIS??.DOC file 21
PARREP - Replace parameter data 21
PARDIS - Display parameter data 21
REP - Modify display parameter 22
SPE - Load new soft keys 23
UC2 - Load new soft keys 23
3.0 CHANGE COMMAND GROUP **24**
UC6 - Change point command 25
UC6REF - Change refresh rate command 25
UC6JI - Jog point using increment of 1 25
UC6JD - Jog point using decrement of 1 25
UC6tbl - Change point directly 25
4.0 ALARM COMMAND GROUP **27**
Enabling/Disabling Alarm(s) 27
5.0 TAGNAME COMMAND GROUP **28**
TAGPRI - Display all loaded tagnames 28
TAGDIS - Show Tagname detail data 28
6.0 TREND COMMAND GROUP **29**
7.0 RECIPE COMMAND GROUP **30**
UC7 - Request Recipe System Action 31
UC70 - Display names 1-60 31
UC71 - Displays next 60 31
UC72 - Displays previous 60 31
UC73 - Rename selected recipe 31
UC75 - Do Action Code 32
UC76 - Select Recipe 32
UC5 - Rename Recipe using Prompts 33
UC1 - Recipe Compare/Edit Functions 34
UC10 - Compare Temp to Active 34
UC11 - Compare Temp to Disk 34

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

UC12 - Go to next page 34
UC13 - Go to previous page 35
UC14 - Expand a super ingredient 35
UC15 - Contract a super-ingredient 35
UC16 - Update Active data 35
UC17 - Edit Temp Ingredient 35
UC18 - Reserved for internal use 35
UC19 - SDCS Compare Temp to Active 35

8.0 DRIVE COMMAND GROUP **38**
UC3 - Send Drive Command(s)
UC4 - Sends the "Tab" register to PLC
UC9 - Drive Draw Selection Commands

9.0 RECORDER COMMAND GROUP **56**
PEN - Control Pen Recorder 57

10.0 MISCELLANEOUS COMMAND GROUP **59**
QUI - Quit command 60
HEL - Help command 61
SWI - Switch window command 62
LOG - Login command 63
LOC - Logout command 64
SUB - Submit the SC300.SUB command file 65
PRT - Print screen to hard copy device 66
FIR - Run another program 67
FOR - Force program to execute 68
UTI - Utility commands 69
MEM - Display free heap memory 70

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

1.0 Introduction

This manual provides a detailed description of the SC-300 command language. Commands are grouped according to the major functional area in which they normally are used. Since some commands are multi-faceted in how they are used, the primary use of the command determines its group placement. Please refer to the table of contents for rapid lookup of a specific major command.

C A U T I O N

Commands not listed in this manual are not supported as part of the SC-300 product and should not be used. Failure to observe this caution *may* result in damage to the machine(s) being monitored or controlled.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

II. Display Command Group

The SC-300 system can support from 1 to 96 dynamic screen displays. Several of these 96 displays are pre-defined for use by the SC-300 software while the others are user-definable. Each display (screen) is composed of two parts --- background data and foreground data. Commands are provided in the SC-300 software to manipulate both aspects of screens.

The background data is the definition of textual data that never changes. This is the data that is drawn once when the display is first called up on the screen. This background data can contain both text and special characters as defined in the character generator EEPROM supplied with the software for your graphics hardware.

The foreground data is the definition of the dynamic data to be displayed on top of the background screen. This definition includes the display representation information, coordinates, color, and security levels.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: DIS [a]

Sub-Commands:

None

Parameters:

a. Display number with a range from 1 to 96

Security Level: 2

Short Description:

Displays a new screen to the operator.

Description:

This command clears the screen, draws the associated background format for the requested display, starts updating the dynamic data for the screen, and loads the primary set of function key definitions for this display from FUNCTION.IBM

If the requested display does not exist, an error message will be displayed and the current display will remain unchanged.

If the parameter is not specified on the same line as DIS, then the system will prompt for the display number using the following prompt: DIS?

Example(s):

?DIS20

Displays screen 20

?DIS 3

Displays screen 3

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: PRE

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

Displays a previous screen to operator

Description:

This command will redraw the previous display requested using the DIS command. This command is especially useful when alternating between an operating screen and perhaps another operating screen.

Example(s):

?PRE

Displays screen drawn prior to the current screen.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: PGU
PGD

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

Pages up or down through the displays.

Description:

These commands allow the user to scroll through the list of displays without knowing their display numbers. The PGU command increments the display number by one while the PGD command decrements the display number by one.

If a display is not found, the current display will continue to be shown.

Example(s):

?PGU

If the current display was 4, then this command would attempt to call up screen 5.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: ESC

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

Simulate ESCAPE key press.

Description:

This command allows the escape key to be simulated from the command line. This command is normally used as part of a function key program to terminate a command string.

Example(s):

?ESC

This command would terminate the current command if one were active.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: SHO [a]

Sub-Commands:

None

Parameters:

a. Format number with a range from 1 to 96

Security Level: 5

Short Description:

Displays a new background format to the operator

Description:

This command clears the screen and draws the associated background format for the requested display.

If the requested format does not exist, an error message will be displayed and the current display will remain unchanged.

If the parameter is not specified on the same line as SHO, then the system will prompt for the display number using the following prompt: FOR?

Example(s):

?SHO20

Draws background data for screen 20

?SHO 3

Draws background data for screen 3

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: ADD

Sub-Commands:

ADDDIS

Parameters:

None.

Security Level: 5

Short Description:

ADD creates the dynamic data definition for a screen.

ADDDIS loads a text file of dynamic data definitions for a new screen.

Description:

ADD

This command will allow interactive specification of all the live data points to be shown on a screen. It searches the background format currently being displayed for question marks. Each question mark will cause a series of prompts to be presented asking for detail data about the dynamic data point.

ADDDIS

This command will read a file named DISnn.DOC created by the PARDOC command. This text file answers all of the questions normally asked interactively when using the ADD command.

If the requested format does not exist, an error message will be displayed and the current display will remain unchanged.

General Notes:

If the requested display already exists, it must first be deleted using the DEL command.

The General Electric Screen Linker utility program should be used to create dynamic definitions for displays since it supports named data points.

A display can contain no more than 128 dynamic data points.

Data entry notes: While entering answers to prompts, several responses are possible. If an escape key is pressed, the command will be aborted. If a semicolon is pressed, the

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

command will be considered complete and an OK? prompt will normally appear. If a Y is answered to this prompt, the command will be executed. If an N is answered, the command will not be executed.

Example(s):

?ADD

This command will prompt for the format number and the display number to create. It then will prompt for detail about each question mark found on the display.

?ADDDIS

This command will prompt for the display number to create. It then will read into memory the DISnn.DOC file where nn is the display number specified.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: DEL [a]

Sub-Commands:

None

Parameters:

a. Display number with a range from 1 to 96

Security Level: 4

Short Description:

Deletes a display file from disk.

Description:

This command will delete the requested display file (DISnn.IBM) from disk.

If the requested display does not exist, an error message will be displayed and the current display will remain unchanged.

If the parameter is not specified on the same line as DEL, then the system will prompt for the display number using the following prompt: DIS?

Example(s):

?DEL20

Deletes the display file named DIS20.IBM from disk.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: REM [a]

Sub-Commands:

None

Parameters:

a. Format number with a range from 1 to 96

Security Level: 5

Short Description:

Deletes a format file from disk.

Description:

This command will delete the requested format file (FORnn.IBM) from disk. It is normally used immediately prior to creating a new format file using the CRE command.

If the requested format does not exist, an error message will be displayed and the current display will remain unchanged.

If the parameter is not specified on the same line as REM, then the system will prompt for the display number using the following prompt: FOR?

This command does not change any display files that may require this deleted format. It is up to the user to delete the associated display files using the DEL command.

Example(s):

?DEL20

Deletes the display file named DIS20.IBM from disk.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: KIL [a]

Sub-Commands:

None

Parameters:

a. Display number with a range from 1 to 96

Security Level: 4

Short Description:

Deletes a display from memory. The disk resident version is not affected.

Description:

This command will delete the requested display from the SC-300 memory. The system buffers in memory the most recently used displays to improve the time required to re-display a screen. When changes are made to the disk resident version of the display, it may be necessary to kill the memory resident version to force the system to reload from disk.

If the parameter is not specified on the same line as KIL, then the system will prompt for the display number using the following prompt: DIS?

This command does not affect the disk resident version of the display and can be used without fear of deleting the display forever.

It is possible to control the size of the system memory buffer used for display foregrounds and backgrounds. The ADDHEAP command can be used to increase or decrease the size of the memory available to the SC300.CMD program.

Example(s):

?KIL20

Deletes the display numbered 20 from the system memory buffer.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: FRE [a]

Sub-Commands:

None

Parameters:

a. Format number with a range from 1 to 96

Security Level: 5

Short Description:

Deletes a format from memory. The disk resident version is not affected.

Description:

This command will delete the requested format from the SC-300 memory. The system buffers in memory the most recently used formats to improve the time required to re-display a screen. When changes are made to the disk resident version of the format, it may be necessary to kill the memory resident version to force the system to reload from disk.

If the parameter is not specified on the same line as FRE, then the system will prompt for the display number using the following prompt: FOR?

This command does not affect the disk resident version of the format and can be used without fear of deleting the format forever.

It is possible to control the size of the system memory buffer used for display foregrounds and backgrounds. The ADDHEAP command can be used to increase or decrease the size of the memory available to the SC300.CMD program.

Example(s):

?FRE20

Deletes the format numbered 20 from the system memory buffer.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: CLE

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

Clears the current display from the screen. The function keys are not affected by this command.

Description:

This command can be used to clear the display screen area prior to using the DRAW command.

Example(s):

?CLE

Clears the screen.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: DRA

Sub-Commands:

None

Parameters:

None

Security Level: 5

Short Description:

Causes the SC-300 program to enter a screen drawing utility to allow specification of the background format data.

Description:

This command causes the system to enter a specialized drawing utility. The screen is NOT cleared prior to entry. This allows modification of an existing format that might already be shown using the SHO command.

The General Electric SCDRAW utility program provides exactly the same functionality in an offline environment and is the preferred technique for building screen formats since it automatically creates the format file.

The DRA command is exited by pressing the escape key. The screen may then be saved as a format file using the CRE command.

Example(s):

?DRA

Enters the draw mode.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: CRE [a]

Sub-Commands:

None

Parameters:

a. Format number to create in the range 1 to 96.

Security Level: 5

Short Description:

Creates a file named FORnn.IBM from the currently displayed information on the screen.

Description:

This command is used following a DRA command and a REM command. It reads the display memory and creates a file of escape sequences to redraw the screen. The file is named FORnn.IBM where nn is the format number.

If the format file already exists, an error message will be displayed to the user.

The CRE command also can create a documentation file or printout that indicates where each question mark is located on the screen. This printout can be used to document the live data points prior to using the ADD command. The documentation output destination is specified using the SC300.CFG text file. Documentation files are named FORnn.DOC where nn is the format number. The default output device is the currently active printer number in use by the virtual console.

The General Electric SCDRAW utility program provides exactly the same functionality in an offline environment and is the preferred technique for building screen formats since it automatically creates the format file immediately after drawing it. This tool does not require the use of the REM or DRA commands from within the SC300 program.

Example(s):

?CRE5

Creates a format file named FOR05.IBM.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: REF

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

Refreshes the dynamic data on the current display

Description:

After using some commands, it is necessary to restart the display refresh process. This command is used to restart the display updating and can also be used to refresh the screen immediately when a long screen update rate is currently defined.

Example(s):

?REF

Restarts the screen updating process and causes an immediate update of all changed dynamic data points.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: PAR

Sub-Commands:

PARADD - Add a new display parameter
PARDEL - Delete a display parameter
PARDOC - Create DIS??.DOC file
PARREP - Replace parameter data
PARDIS - Display parameters serially

Parameters:

None

Security Level: 5

Short Description:

Allows the addition, deletion, display, replacement, and documenting of display parameters.

Description:

These parameter commands allow the user to manipulate individual display parameters for the current display screen. Parameters can be added to an existing display, deleted from the display, or modified. In all cases, a series of prompts are used to determine which parameter is to be used along with any supporting information.

The General Electric Screen Linker utility program is the preferred method for manipulating display parameters and the use of these PAR commands should be avoided.

Data entry notes: While entering answers to prompts, several responses are possible. If an escape key is pressed, the command will be aborted. If a semicolon is pressed, the command will be considered complete and an OK? prompt will normally appear. If a Y is answered to this prompt, the command will be executed. If an N is answered, the command will not be executed.

PARADD This command is used to add a new display parameter to an existing display. The user is requested to enter the parameter number, XY coordinates, database type, database index, color, symbol, security, and changeable status. This command does not utilize tagnames to specify a point and it does not update the corresponding SRN file for this display.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

PARDEL This command will remove an existing display parameter from the current screen. The DIS??.IBM file will be modified to reflect the deletion. The corresponding SRN file will not be updated.

PARDOC This command can be used to create a text file that lists each and every parameter defined on the current screen. This command is normally used in conjunction with the ADDDIS command.

PARREP This command allows the user to change one of the settings for a pre-defined screen parameter. Items that can be modified include color, security, changeable status, and database point. Changes made using this command are NOT saved in the DIS??.IBM file. Use the REP command to store the changes permanently.

PARDIS This command is used to display the values of each display parameter. The user is shown the first parameter. When a semi-colon is pressed, the next parameter will be displayed. This sequence continues until an escape key is hit.

Example(s):

?PARDOC

Creates a documentation file named DISnn.DOC for the current display.

?PARADD

Allows the user to specify a new live data point on the current display. The cursor keys can be used to identify the parameter's XY coordinates after invoking the command.

?PARREP

Modifies the memory resident version of the display parameter. Changes to the parameter attributes are NOT saved in the corresponding DISnn.IBM file. Use the REP command to modify both the memory and disk resident files.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: REP

Sub-Commands:

None

Parameters:

None

Security Level: 5

Short Description:

Allows modification of a display parameter and saves the changes on disk in the DISnn.IBM file.

Description:

This command is used to change one or more attributes of a single display parameter. After terminating all the changes by pressing the semi-colon, the changes will be saved on disk and in memory.

Pressing the escape key will abort the REP command without making any changes.

Data entry notes: While entering answers to prompts, several responses are possible. If an escape key is pressed, the command will be aborted. If a semicolon is pressed, the command will be considered complete and an OK? prompt will normally appear. If a Y is answered to this prompt, the command will be executed. If an N is answered, the command will not be executed.

Example(s):

?REP

Begins the prompt sequence for changing attributes of a single display parameter.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: SPE [a]
UC2 [a]

Sub-Commands:

None

Parameters:

a. Function key menu number from -200 to 200

Security Level: None

Short Description:

Loads a new set of function key definitions from one of three data files---FUNCTION.IBM, SPECIALS.IBM, or DRIVES.IBM.

Description:

The SC-300 program obtains function key programming data from one of three files. The FUNCTION.IBM file contains the primary function key program for each display. It contains exactly 96 sets of function keys. The SPECIALS.IBM file contains the secondary function key programs for each display PLUS additional menus as required by the application. This file can hold up to 200 sets of function key programs.

The DRIVES.IBM file contains the programming for function keys for drive commands. It is invoked by loading the primary menu for display number -101 or greater (a non-existant display).

The SPE command is used to load function key programs from one of these sources. The parameter is the menu number to load.

When the parameter is 1 to 200, SPECIALS.IBM is used.
When the parameter is -1 to -100, FUNCTION.IBM is used.
When the parameter is -101 thru -200, DRIVES.IBM is used.

The actual function key programming is performed using the General Electric Function Key editor utility program.

The UC2 command is a synonym for SPE and can also be used to load new soft keys. The preferred command for new applications is SPE.

Example(s):

?SPE 35

Loads the function key program from SPECIALS.IBM for display 35.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

III. Change Command Group

The commands in this group are used to modify the values of points in the database. These commands allow incrementing and decrementing values as well as directly specifying a new value. In addition, a command to adjust the refresh rate of the screen is included.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: UC6

Sub-Commands:

UC6REF [a]
UC6EUIbbbb*ccccc
UC6EURbbbb*cccccc
UC6FLGbbbb*c
UC6DIIbbbb*c
UC6DOObbbb*c
UC6SCAbbbb*ccccc
UC6OUTbbbb*ccccc
UC6JI
UC6JD

Parameters:

a. Refresh rate in seconds
b. Database index
c. New value of database point

Security Level: 3

Short Description:

Used to change the value of a highlited display parameter. Can also be used to change the screen refresh rate.

Description:

The UC6 command is used to change values in the database. When used with no parameters, it directs the value change to the currently highlited display point on the screen. If no point is highlited, an error message will be displayed to the user. The security level of the point is used to enable/disable changes using this command in addition to having a command level security high enough to use the UC6 command.

When used in this fashion, the command will invoke a special set of function keys that program F1 and F2 to "Accept" and "Reject" the entered value. The user can either press F1 or F2 to complete data entry, press return to get an "OK?" prompt, or press escape to abort.

The UC6REF command can be used to change the display refresh rate. Values from 1 to 99 can be specified.

The UC6xxx commands where xxx is one of the valid database types can be used to change a value in the database that may not be shown on the current display. When used from the command line, the user is always prompted for the new value. In this case, the asterisk should NOT be used in the command string. When used to

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

program a function key, the asterisk can be used to separate the database ID from the new value. In this case, a single function key can change the value of any point in the database to a pre-defined value. The point being changed has no associated security level when using either of these two techniques.

The F1 and F2 function keys operate in a similar fashion to the plain vanilla UC6 command with no parameters.

The UC6JI and UC6JD commands perform an increment/decrement function on the currently highlited display parameter. The increment/decrement value is 1.0. The value change is clamped by the high or low data entry limits specified for the point. The actual value sent to the Series Six is the final value when the jog function key is released. Point level security is active when using these two commands.

Example(s):

?UC6

Displays the current value of the highlited point and asks for the new value. After typing the new value the user can press F1 to accept entry or press Enter twice to confirm entry.

?UC6REF

Prompts the user for the new display refresh rate in seconds.

?UC6EUI349

Displays the current value of database point EUI 349 and asks for a new value. Entry is the same as the UC6 command.

?UC6EUR349*123.5

Can only be used to program a function key. Performs an absolute change on database point EUR 348 giving it a value of 123.5. No user interaction is required when using this form of the command.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

IV. Alarm Command Group

There are relatively few commands that are available to the user while on-line. The only commands available include enabling and disabling a pre-configured alarm. These functions are accomplished using the UC6EUIxxxx command.

In general, the function key menu system is programmed to direct a change to database point EUI139. When an alarm is to be enabled/disabled, the alarm number is placed into this database point. The alarm system will monitor this location and perform the action required. If the alarm is currently enabled, then it is disabled. If the alarm is disabled, it is re-enabled.

If an english language alarm is configured for this point (EUI139) with a high limit of +0.5, then a message will be generated whenever an alarm has been re-configured. (This assumes that Setpoint logging has been enabled as a feature).

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

V. Tagname Command Group

Tagnames are used to associate an english description with an actual database point location. A tagname can be from 1 to 9 characters in length. If the point is an analog point, an additional 24 characters of text can be used to describe the point. In addition, a six character string is used to indicate the engineering units used by the point.

If the point is digital, an additional 18 characters of text are used to describe the point. In addition, the SET and RESET states are described by two 6 character descriptions.

The pen recorder function of the SC300 program requires entry of tagnames to specify the point to be recorded. In order to assist the user in determining the spelling, a display of all memory resident tagnames can be requested using the TAGPRI command.

The detail data for any specific tagname can be shown by using the TAGDIS command. Use of the TAGDIS command is normally not required and is available only for system development.

Tags are configured using the General Electric System Configuration utilities. Tag reconfiguration while on-line is not possible with the SC-300.

Data entry notes: While entering answers to prompts, several responses are possible. If an escape key is pressed, the command will be aborted. If a semicolon is pressed, the command will be considered complete and an OK? prompt will normally appear. If a Y is answered to this prompt, the command will be executed. If an N is answered, the command will not be executed.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

VI. Trend Command Group

The use of the short term trends capability within the SC-300 product is currently not supported.

Available unsupported commands include:

TREDIS - Shows Trend detail data for all trends
TREDOC - Creates text file listing all defined trends
TREDEL - Deletes defined trend from TREND.IBM and memory
TREADD - Adds new trend to TREND.IBM and memory
TRELOD - Performed Automatically at LOGGER Startup
TREREP - Replaces trend parameters with new parameters

Data entry notes: While entering answers to prompts, several responses are possible. If an escape key is pressed, the command will be aborted. If a semicolon is pressed, the command will be considered complete and an OK? prompt will normally appear. If a Y is answered to this prompt, the command will be executed. If an N is answered, the command will not be executed.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

VII. Recipe Command Group

These commands are associated with the advanced recipe system provided in the SC-300 system. Commands to upload, download, edit, compare, print, and name recipes are provided.

The recipe commands deal with recipes that can be located in one of three places---Active, Temp, or Disk. Active recipes are assumed to be stored in the Series Six programmable controller. Temp recipes are stored in a special file named SCRATCH.RCP on the currently defined ram-drive. Disk recipes are stored on the hard disk in the recipe storage user area. There can only be one SCRATCH.RCP while there can be up to 32 active recipes and many more disk recipes.

The recipe system allows defining a group of ingredients into a CLASS. The system supports from 1 to 32 classes. Each class has its own list of ingredients and data files. An ingredient can belong to more than one class. A class can hold from 1 to 4000 ingredients or super-ingredients.

The recipe system also supports a concept called SUPER-INGREDIENTS. A super ingredient has no actual value. It is merely a list of ingredients. A super-ingredient may contain from 1 to 20 ingredients or super-ingredients. Super-ingredients may be nested up to four levels deep.

Super ingredients allow the system designer to organize the display of ingredient data on the recipe editing screens. Less frequently accessed ingredients should be grouped into a super-ingredient to allow placing more important ingredients on the first display page.

Each class can have an associated report file with a filetype of RPT. This file can contain many different report generation commands to allow customized printing. This report file is used to print both the active recipe values as well as the disk resident recipe values.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: UC7

Sub-Commands:

UC70 - Display first 60 recipe names
UC71 - Displays next 60 recipe names
UC72 - Displays previous 60 recipe names
UC73 - Rename selected recipe
UC74 - Not implemented
UC75 - Perform recipe action per code
UC76 - Select Recipe
UC77 - Reserved for internal use by compare function

Parameters:

Recipe action codes are as follows:

1 - XFR Temp to Active
2 - XFR Active to Temp
3 - XFR Disk to Active
4 - XFR Active to Disk
5 - Print from Disk
6 - Print from Active
7 - XFR Disk to Temp
8 - XFR Temp to Disk

Security Level: None

Short Description:

Description:

UC70 This command will cause the recipe name screen to be displayed (#88) with the first 60 recipe names defined for the desired class. The class can be specified as part of the command or the system will prompt for it. The system will attempt to guess which class you want and allow the user to overide its selection. For example, this command could be entered as either UC70 or UC70TDR where TDR is the desired class name.

UC71 This command will display the next 60 recipe names. The command will wrap around to the first 60 if there are no more names to display. The total number of recipes is determined at system initialization by reading the SC300.CFG file for the maximum recipe number allowed.

UC72 Similar to UC71 except it goes backward by 60 names.

UC73 This command will allow the user to rename a recipe that is

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

already named. Blank recipes can not be named using this command.

UC75 This command will perform the recipe action on the selected recipe. For example, to execute the Temp recipe to Active the following command would be given: UC751

The action codes are listed above. When a recipe name is required, the user can point to it and then use the UC76 command to select it. The UC75 command will then complete its action.

This command can be invoked from any display in the system and will prompt for any missing information.

UC76 This command is used to select recipes when a selection is required. Selections are required for naming, and performing an action code. If the recipe name screen was called up from the compare system, this command will return back to the compare screen with the selected disk recipe.

Example(s):

See above.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: UC5

Sub-Commands:

Parameters:

An optional class name can be specified.

Security Level: 2

Short Description:

This command allows renaming a recipe when the recipe number and class are known.

Description:

This command can be used to rename a recipe when the user prefers to number recipes. In this case, the class name and the number are prompted prior to asking for the new name. A name can be from 1 to 62 characters in length.

If there is only one class, it is possible to provide the class name as part of the command (UC5GRM) to eliminate the system asking for the class each time this command is invoked.

The entered recipe number must be in the range from 1 to the maximum allowed in the SC300.CFG configuration file. No more than 999 recipes per class are allowed under any cirmcumstances.

Recipe 0 is reserved for use by the recipe system.

The recipe names are stored in the file named using the following conventions: SC-aaa.NAM where *aaa* is the three letter class name.

Example(s):

?UC5

This will prompt the user for the class and recipe number to name. Any existing name will be overwritten.

?UC5GRM

This version will only ask for the recipe number. Other action is identical to the first example.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: UC1

Sub-Commands:

UC10 - Compare Temp to Active
UC11 - Compare Temp to Disk
UC12 - Go to next page of compare screen
UC13 - Go to previous page of compare screen
UC14 - Expand a super ingredient
UC15 - Contract a previously expanded super-ingredient
UC16 - Update Active data shown on compare screen
UC17 - Edit Temp (normal ingredient only)
UC18 - Reserved for internal use by SC-300
UC19 - Compare Temp to Active from SDCS Screen Menus

Parameters:

None

Security Level: None

Short Description:

These commands allow comparing the Temp recipe with either the active or disk resident recipe files. In addition, editting of the Temp recipe is performed using these commands. Display 89 is used exclusively by these commands.

Description:

UC10 This command is used to compare the values contained in the temp file with those in the active recipe file. The class of Temp is used to determine the recipe class. If display 89 is not displayed, it will automatically be displayed.

UC11 This command is used to compare the values contained in the temp file with those in the disk resident file(s). The class of Temp is used to determine the recipe class. If display 89 is not displayed, it will automatically be displayed. The user is prompted for the name of the recipe on disk. Several data entry techniques are available. The user may type the first few unique characters of the name and the system will show each name match starting with column 1. The user may accept or reject the computer's selection. Alternatively, if the name string is preceded by an asterisk (*), every recipe name that contains the entered string will be selected for confirmation. If the name is not known, the user may type a '?' <cr> and the system will then present the first screen of recipe names. The user selects the recipe name (UC76) and then this command will continue.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

UC12 This command will redisplay screen 89 showing the next 20 ingredients or super-ingredients. If no more pages are available, the first screen of ingredients are displayed.

UC13 Similar to UC12 except it goes back one screen.

UC14 This command will expect that the tab key is pointing to a super-ingredient. If so, the command will re-display the contents of the super-ingredient.

UC15 This command will return one level back from the current super-ingredient expansion.

UC16 This command provides a method to update the active data shown on the compare to active display. Normally, the active data is not updated at fixed intervals and must be updated manually.

UC17 This command will allow the user to change the value of an ingredient in the Temp recipe area. Only normal ingredients can be editted. The user's security must be at a level equal to or greater than the ingredient security. Modifying the recipe will cause the display to indicate that the recipe has been changed since it was first captured to Temp.

UC18 This command is reserved for internal use by the SC-300 system.

UC19 - SDCS Compare TEMP RECIPE to ACTIVE SDCS DATA

The UC19 is a special subset of the UC1 commands and have additional parameters to describe to the system further information as to what the operator wants. It is very important that this command NOT BE TYPED AT THE ACTIVE SC300 CONSOLE unless on levels 6 or 8 in SDCS (DRIVES.IBM function keys). It is also important to include the UC19 command (without any other parameters) immediately following the command to go to level 6 or 8 in DRIVES.IBM.

UC19 is only to be used with the SUPER DRIVE CONTROL SCREEN (SDCS). It is highly customized to operate within the confines of the SDCS system within the SC300 system. There are linkages to the UC3 command language with UC19 that will be explained later.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Sub-Command: UC19

There are five different flavors of the UC19 and a sixth type which is actually a UC3 command. The first type of UC19 command has no parameters included and must be invoked once function key levels 6 or 8 (from DRIVES.IBM) has been called to the SC300 screen. This command simply reads the TAB register (SEE UC3 for an explanation on the TAB register), and determines in which drive the operator is interested. It retrieves the setpoints in the TEMP recipe and puts those setpoints the system designer has linked to that drive on the function key line on level 8. If the operator is on level 6, the command will retrieve the mode word from the TEMP recipe, and decode the bits to highlight on the function keys which regulator was active at the time the TEMP recipe was saved. In both cases, the TEMP recipe must have been loaded from the ACTIVE database or from DISK prior to issuing this UC19 command. If TEMP is empty, an error message is displayed, and any subsequent data on the function keys are meaningless until a TEMP recipe is loaded. In the case of level 6, the mode word MUST be an ingredient in the class definition for the recipe in TEMP in order for the information on the level 6 keys to be meaningful.

Sub-Command: UC19Ex x

The second command applies to level 6 only. The UC19 command is followed with the parameter 'E' (UC19E). This command allows the changing of the active regulator in the TEMP recipe. For example, if the operator sees that the active regulator in the TEMP recipe is DRAW and he requires when he transfers the TEMP recipe to ACTIVE that the active regulator be CURRENT, he may use this command to make the CURRENT regulator the active regulator in the TEMP recipe. Once he does this, the regulator of the drive, when TEMP is transferred to ACTIVE, will change from whatever it is to CURRENT. The additional parameters actually carry out the desired command when the TEMP recipe is transferred to ACTIVE and utilizes the STATUS-S and UC3CM style of commands.

Remember, that only regulator changes may be made on level 6, therefore; the only Status-S command words available to do this are Command Word 0 and Command Word 4, and only those bits relating to regulator changes in those two words are allowable. (SEE UC3 for bit descriptions in Status-S command words). It is important that this command only be executed from level 6 function keys and from the function key itself (not directly typed into the SC300 console).

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Sub-Command: UC19U

This sub-command is a hidden command and should not be used by the system designer on any function key. This command is utilized internaly to SC300 and writes any changes made on level 6 or level 8 into the TEMP recipe.

Sub-Command: UC3EX SE

Although this command is in the UC3 variety it can only be used on function key level 8 (DRIVES.IBM). This command allows the selection and editing of a setpoint in the TEMP recipe. When this command is invoked, a setpoint must be highlighted on the function key level 8. This is the setpoint that will be edited. SC300 will then show the present value of that setpoint (TEMP recipe setpoint) and ask for a new number. The number entered is limit checked under the same rules as editing an active setpoint. SC300 asks for confirmation of the change and with the use of the UC19U command, writes the new number to the TEMP recipe.

Sub-Command: UC19I

This command increments the selected setpoint on level 8 by a whole number (1.0) without asking for confirmation. This command must be used under a function key of type JOG in order to function properly.

Sub-Command: UC19D

This Command decrements the selected setpoint on level 8 by 1.0, again without asking confirmation from the operator. Like UC19I, it must be used only on function keys and that function key must be defined as type JOG. Both UC19I and UC19D use the command UC19U to update the TEMP recipe once the operator releases the function key to stop the increment or decrement. Both these functions utilize the same limit checking as incrementing or decrementing an active regulator setpoint.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

VIII. Drive Command Group

These commands are normally programmed within the DRIVES.IBM function key programs. These commands allow specialized memory changes to take place on certain database values. In general, the current display must be one of the Drive Displays (60 thru 87). The hilighted parameter on the screen is associated with a specific drive, zone, or line. Depending on the currently highlited parameter, the drive command performs one or more actions for the drive, zone, or line.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: UC3

Sub-Commands:

UC3CMx y - Status-S Bit Command
UC3SET cc - SDCS Setpoint Change Command
UC3CMZ c - SDCS Active Regulator Setpoint Change Command
UC39EX SE - SEE UC1

Parameters:

UC3CM Sub Command
x - Status-S word identifier
y - Bit identifier in Status-S word

UC3SET Sub Command
cc - Selects the setpoint and tells the program how to handle it

Security Level: None

Short Description:

UC3 commands are associated with the operation and maintenance of drives under the SC300 system's control. They allow the control of state changes of a drive or a group of drives. They also allow operators to send setpoint changes. All commands are geared to the control and operation of an entire coordinated line through a special LAN protocol for the DC300 drive system called Status-S. This protocol requires interfacing software in the Series Six Programmable Controller and is not meant to be a stand-alone communication protocol directly to the drives.

Description:

UC3 Commands are highly custom commands and have many restrictions applied to them. These commands are not normally expected in the typed form and are most commonly used in association with the function keys. There is a special set of function keys related to the Super Drive Control Screen (SDCS) and is contained in the DRIVES.IBM file. There are seventeen (17) levels available in DRIVES.IBM and each level has been assigned a special purpose. These levels contain all the necessary UC3 commands to allow the operation of the drives in any given system. In addition to the commands, each level of DRIVES.IBM gives the program additional information on how to handle each particular command. In some cases, the program requires two commands to be executed consecutively to allow proper actions to take place.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

When using this command reference, the UC3 command structure requires careful consideration of what each command will do and to which group of drives it will do it to. In particular, one should exercise care when directly typing these commands on an active SC300 console as most of the commands are meant to be used solely with the function keys. This command reference manual will describe the commands in the context they are used with the function keys and not as directly typed commands.

First, a few words about Status-S. Status-S is a method of communicationg commands and setpoints to the drive as well as a method for the drives to report its feedbacks and status. Status-S is handled as a special memory map inside the Series Six BIU card (SuP) and the Micro-Application Card or LAN card (MuP) on each drive. Special parameters must be programmed in each drive to allow Status-S to be understood at the drive. SuP requires special handling through the Series Six DPREQ windows and special sequencing/initialization to operate with Status-S properly. There are four catagories of data communicated between the drives and the Series Six. They are:

1) Bits to the Drive (Commands)
2) Variables to the Drive (Setpoints)
3) Bits from the Drive (Status)
4) Variables from the Drive (Feedbacks)

The definition of these bits and variables are too large to describe here, however; each catagory has many, many words of possible data that may be communicated. It is up to the system designer to pick out of the maximum case which bits and variables are important to any particular drive in any particular system with a single exception; the bits to the drives are always in the max case form. These bits are controlled by the UC3CM sub-commands and the tab register. The tab register is a special variable created by SC300 to inform the Series Six to which drive(s) the command must go. A description of the tab register and the bits to the drives are contained in this manual on subsequent pages. Variables to the drives are controlled through the use of UC3SET sub-commands. If the system designer has not picked a certain setpoint to be communicated to the drives, it is impossible for SC300 to send this setpoint. SC300 anticipates this kind of error and will inform the operator that the setpoint is not available to send. Again, the tab register is used to designate which drive(s) are to receive the setpoint being altered.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

The TAB register is created within the SDCS system and relies heavily upon the particular function key level the operator is currently on. There are several ways of selecting a drive or a group of drives to send commands to. The register is separated into three parts with the first seven bits used to show the drive's drop number that is currently highlited on the SC300 screen, the next five bits show which zone that drop belongs to, and the last four show what "group" mode the tab is in. There are three modes: LINE, ZONE, and DRIVE and two more qualifiers that further describe LINE and ZONE modes: PROGRESSIVE and REGRESSIVE. The breakdown of the tab register is shown below. It is with this exact tab register that SuP knows which drives are to receive the commands that are along with it in the memory map.

TAB REGISTER DESCRIPTION

```
  F    E    D    C    B    A    9    8    7    6    5    4    3    2    1    0
  |    |    |    |    |    |    |    |    |    |    |    |    |    |    |    |
  X    X    X    X    X    X    X    X    X    X    X    X    X    X    X    X
  |    |    |    |    |____|____|____|____|    |____|____|____|____|____|____|
REGRS  |  ZONE   |              |                          |
  |    |  COMND  |       ZONE ID (0 - 31)                DROP ID (0 - 93)
  |  PROGR  |  LINE    ZONE # IS THE ZONE OF      DROP # IDENTIFIES  WHICH  SUP  &
  |    |    |  COMND   THE CHANNEL  IDENTIFIED      WHICH  CHANNEL  OF THAT  SUP
  |    |    |    |         BY THE DROP #          (00  -  31 = 1ST  SUP,  CHANNEL  P)
  |    |    |    |                                (32  -  63 = 1ST  SUP,  CHANNEL  Q)
  |    |    |    |                                (64  -  95 = 2ND  SUP,  CHANNEL  P)
  |    |    |    |                                (96  - 127 = 2ND  SUP,  CHANNEL  Q)
  |    |    |    |
  |    |    |    |        ************************************************
  |    |    |    |        * NOTE : SC300  CAN  HANDLE  93 DRIVES  MAX            *
  |    |    |    |        *         DROPS  00,  32,  64,  96 ARE  RESERVED  FOR  SUP *
  |    |    |    |        * EXAMPLE  :  TOTAL  DROPS  AVAILABLE          =   127 *
  |    |    |    |        *             TOTAL  DRIVES                    =  - 93 *
  |    |    |    |        *             TOTAL  CHANNELS                  =  -  4 *
  |    |    |    |        *             ---------------------------------      *
  |    |    |    |        *             DROPS  AVAILABLE  FOR  DROP-INS   =    31 *
  |    |    |    |        ************************************************
  |    |    |    |
  0    0    0    0  = DIRECTED  TO A SINGLE  DRIVE
  0    1    0    1  = BROADCASTED  FROM  PIVOT  TO THE "END  OF  LINE"  (CONFIGURABLE)
  1    0    0    1  = BROADCASTED  FROM  PIVOT  TO THE "START  OF  LINE"
  0    0    1    0  = ZONE  SEND  TO ALL DRIVES  IN THE SPECIFIED  ZONE
  0    1    1    0  = ZONE  SEND  FROM  PIVOT  TO THE "END  OF  ZONE"  IN THE SPECIFIED  ZONE
  1    0    1    0  = ZONE  SEND  FROM  PIVOT  TO THE "START  OF  ZONE"  IN THE SPECIFIED  ZONE
```

The Bits to the Drive are organized in a contracted area of SC300 and the Series Six. These bits are communicated directly to the drives from SuP upon the changing of state of any bit in any of these words.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

There are currently 19 words of bits reserved in the memory map of which 17 are defined. The word identifier used in the UC3CM sub-command is a single character starting at '0', going up to '9' and then 'A', 'B', and so on until 'H' which is the last word defined. As more bits are defined for different applications, the unused two words 'I', 'J' will be used.

The bit identifier is again a single character starting at '0' for the least significant bit and 'F' for the most significant bit. These bit words are formatted such that setting the bits in the low byte will request that function at the drive; setting the bits in the upper byte of the word will request that that function be toggled to the other state or "turn the function off".

The first word of bits (ACTION_0) is an exception to this rule. The drive will request the function that each bit represents throughout all 16 bits of the word. The first three words of bits will cause operating actions at the drives and will be permissive checked by the Series Six and SC300. The rest of the words of bits are, in general, setup bits that do not cause direct operating actions at the drive. These bits are not permissive checked.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

The Status-S Bits are formatted as follows :

ACTION_0_A (COMMAND WORD 0)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	EXT ON		CUR OFF		PRO ON		DRW ON		CSTOP		STOP		SLOW FWD		RUN FWD

EXT OFF	PRO OFF	ABORT REC	CUR ON	RESET	QSTOP	SLOW REV	RUN REV

ACTION_0_B (COMMAND WORD 1)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	RELEASE SETUPS		JOG FWD STP		RSRVD		RSRVD		DELAY SETUPS		JOG FWD		RSRVD		RSRVD

RSRVD	JOG REV STP	RSRVD	RSRVD	HISTORY LOCK	JOG REV	RSRVD	RSRVD

ACTION_1_A (COMMAND WORD 2)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	SLK TAK UP STOP		PRO INC STP		IRAT INC STP		DRW INC STP		SLK TAK UP		PRO INC		IRAT INC		DRW INC

SLK TAK DN STOP	PRO DEC STP	IRAT DEC STP	DRW DEC STP	SLK TAK DN	PRO DEC	IRAT DEC	DRW DEC

ACTION_1_B (COMMAND WORD 3)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	NO TEN		RSRVD		INDEPENDT		BUMPLESS		TEN AVAIL		DRW INH		COORDINAT		SMOOTH

BIAS DRP ON	TEN CUR	TENSION OFF	DRV ON	BIAS DRP OFF	TEN PROC	TENSION ON	DRV OFF

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

RuP_IN_0_A (COMMAND WORD 4)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	RUN1 CLEAR		SLOW CLEAR		RUN CLEAR		JOG CLEAR		RUN1 SET		SLOW SET		RUN SET		JOG SET
RUN2 CLEAR		RUN0 CLEAR		RUNR CLEAR		JOGR CLEAR		RUN2 SET		RUN0 SET		RUNR SET		JOGR SET	

RuP_IN_0_B (COMMAND WORD 5)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	HOLD DIAMETER CLEAR		MOTOR1 REQUEST CLEAR		COAST STOP CLEAR		REFR INHIBIT CLEAR		HOLD DIAMETER SET		MOTOR1 REQUEST SET		COAST STOP SET		REFR INHIBIT SET
COMMAND FLD ECON CLEAR		MOTOR2 REQUEST CLEAR		LOW GEAR CLEAR		QUICK STOP CLEAR		COMMAND FLD ECON SET		MOTOR2 REQUEST SET		LOW GEAR SET		QUICK STOP SET	

RuP_IN_1_A (COMMAND WORD 6)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	RECAL I LIMIT CLEAR		RECAL REFR CLEAR		INC SPEED STOP		COMMAND FLDM2 CLEAR		RECAL I LIMIT SET		RECAL REFR SET		INC SPEED START		COMMAND FLDM2 SET
SOFT RESET TOGGLE		RECAL TACH CLEAR		DEC SPEED STOP		COMMAND FLDM3 CLEAR		SOFT RESET TOGGLE		RECAL TACH SET		DEC SPEED START		COMMAND FLDM3 SET	

RuP_IN_1_B (COMMAND WORD 7)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
	SPARE		REV REF INHIBIT CLEAR		AUTO REF CLEAR		CURRENT FOLLOW CLEAR		SPARE		REV REF INHIBIT SET		AUTO REF SET		CURRENT FOLLOW SET
SPARE		REGEN INHIBIT CLEAR		FWD REF INHIBIT CLEAR		RUN POSITION CLEAR		SPARE		REGEN INHIBIT SET		FWD REF INHIBIT SET		RUN POSITION SET	

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

RuP_EXTRA_A (COMMAND WORD 8)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
SPARE 06 CLEAR		POS FDB SEL CLEAR		BREAK AWAY CLEAR		PRESET DIAM CLEAR		SPARE 06		POD FDB SEL SET		BREAK AWAY SET		PRESET DIAM SET	
SPARE 07 CLEAR		SPARE 05 CLEAR		XRAMP JR SEL CLEAR		RESCALE TENSION CLEAR		SPARE 07 SET		SPARE 05 SET		XRAMP JR SEL SET		RESCALE TENSION SET	

RuP_EXTRA_B (COMMAND WORD 9)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
SPARE 14 CLEAR		SPARE 12 CLEAR		SPARE 10 CLEAR		SPARE 08 CLEAR		SPARE 14		SPARE 12 SET		SPARE 10 SET		SPARE 08 SET	
SPARE 15 CLEAR		SPARE 13 CLEAR		SPARE 11 CLEAR		SPARE 09 CLEAR		SPARE 15 SET		SPARE 13 SET		SPARE 11 SET		SPARE 09 SET	

SPARE (COMMAND WORD A)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
SPARE 06 CLEAR		SPARE 04 CLEAR		SPARE 02 CLEAR		SPARE 00 CLEAR		SPARE 06		SPARE 04 SET		SPARE 02 SET		SPARE 00 SET	
SPARE 07 CLEAR		SPARE 05 CLEAR		SPARE 03 CLEAR		SPARE 01 CLEAR		SPARE 07 SET		SPARE 05 SET		SPARE 03 SET		SPARE 01 SET	

SPARE (COMMAND WORD B)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
SPARE 14 CLEAR		SPARE 12 CLEAR		SPARE 10 CLEAR		SPARE 08 CLEAR		SPARE 14 SET		SPARE 12 SET		SPARE 10 SET		SPARE 08 SET	
SPARE 15 CLEAR		SPARE 13 CLEAR		SPARE 11 CLEAR		SPARE 09 CLEAR		SPARE 15 SET		SPARE 13 SET		SPARE 11 SET		SPARE 09 SET	

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

LuP_OUT_0_A (COMMAND WORD C)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
LuP_OUT 06 CLEAR	LuP_OUT 04 CLEAR	LuP_OUT 02 CLEAR	LuP_OUT 00 CLEAR	LuP_OUT 06	LuP_OUT 04 SET	LuP_OUT 02 SET	LuP_OUT 00 SET
LuP_OUT 07 CLEAR	LuP_OUT 05 CLEAR	LuP_OUT 03 CLEAR	LuP_OUT 01 CLEAR	LuP_OUT 07 SET	LuP_OUT 05 SET	LuP_OUT 03 SET	LuP_OUT 01 SET

LuP_OUT_0_B (COMMAND WORD D)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
LuP_OUT 14 CLEAR	LuP_OUT 12 CLEAR	LuP_OUT 10 CLEAR	LuP_OUT 08 CLEAR	LuP_OUT 14	LuP_OUT 12 SET	LuP_OUT 10 SET	LuP_OUT 08 SET
LuP_OUT 15 CLEAR	LuP_OUT 13 CLEAR	LuP_OUT 11 CLEAR	LuP_OUT 09 CLEAR	LuP_OUT 15 SET	LuP_OUT 13 SET	LuP_OUT 11 SET	LuP_OUT 09 SET

DROPIN_OUT_0_A (COMMAND WORD E)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
DuP_OUT 06 CLEAR	DuP_OUT 04 CLEAR	DuP_OUT 02 CLEAR	DuP_OUT 00 CLEAR	DuP_OUT 06	DuP_OUT 04 SET	DuP_OUT 02 SET	DuP_OUT 00 SET
DuP_OUT 07 CLEAR	DuP_OUT 05 CLEAR	DuP_OUT 03 CLEAR	DuP_OUT 01 CLEAR	DuP_OUT 07 SET	DuP_OUT 05 SET	DuP_OUT 03 SET	DuP_OUT 01 SET

DROPIN_OUT_0_B (COMMAND WORD F)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
DuP_OUT 14 CLEAR	DuP_OUT 12 CLEAR	DuP_OUT 10 CLEAR	DuP_OUT 08 CLEAR	DuP_OUT 14	DuP_OUT 12 SET	DuP_OUT 10 SET	DuP_OUT 08 SET
DuP_OUT 15 CLEAR	DuP_OUT 13 CLEAR	DuP_OUT 11 CLEAR	DuP_OUT 09 CLEAR	DuP_OUT 15 SET	DuP_OUT 13 SET	DuP_OUT 11 SET	DuP_OUT 09 SET

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

DROPIN_OUT_1_A (COMMAND WORD G)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
DuP_OUT 22 CLEAR	DuP_OUT 20 CLEAR	DuP_OUT 18 CLEAR	DuP_OUT 16 CLEAR	DuP_OUT 22	DuP_OUT 20 SET	DuP_OUT 18 SET	DuP_OUT 16 SET
DuP_OUT 23 CLEAR	DuP_OUT 21 CLEAR	DuP_OUT 19 CLEAR	DuP_OUT 17 CLEAR	DuP_OUT 23 SET	DuP_OUT 21 SET	DuP_OUT 19 SET	DuP_OUT 17 SET

DROPIN_OUT_1_B (COMMAND WORD H)

F	E	D	C	B	A	9	8	7	6	5	4	3	2	1	0
X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
DuP_OUT 30 CLEAR	DuP_OUT 28 CLEAR	DuP_OUT 26 CLEAR	DuP_OUT 24 CLEAR	DuP_OUT 30	DuP_OUT 28 SET	DuP_OUT 26 SET	DuP_OUT 24 SET
DuP_OUT 31 CLEAR	DuP_OUT 29 CLEAR	DuP_OUT 27 CLEAR	DuP_OUT 25 CLEAR	DuP_OUT 31 SET	DuP_OUT 29 SET	DuP_OUT 27 SET	DuP_OUT 25 SET

SPARE (COMMAND WORD I)

SPARE (COMMAND WORD J)

As an example, the SC300 command to ask a drive or a group of drives to run forward would be UC3CM0 0. This would ask SuP to set bit 0 (RUN FORWARD) in word 0 (ACTION 0). Note that there is a space between the word identifier and the bit identifier. This space is required. Natuarally, the SC300 screen must show the name of the drive desired to run as being highlighted with a brown background and bright white foreground color.

To run a group of drives the operator must be on a menu level to OPERATE ZONE or OPERATE LINE. On these menu levels, he is now selecting at least a zone of drives to run. On the LINE OPERATE level, by using the PROGRESSIVE or REGRESSIVE keys, he may select more than one group. On the ZONE OPERATE level, he may select the entire zone or part of a zone. Drives that are in the same zone are shown with a bright white foreground color in the name column on the SDCS screen in SC300.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

More examples

UC3CM4 5 - Sets RUN0 in RuP_IN_0_A (COMMAND WORD 4)
UC3CMC A - Cears LuP_OUT 2 bit IN LuP_OUT_0_A (COMMAND WORD C)
UC3CM3 0 - Sets SMOOTH (regulator transfer) in ACTION_1_B
(COMMAND WORD 3)

Remember that some of these bit commands rely on the function key level itself to provide the appropriate information to SC300 to carry out these commands properly. For example, all commands that are INC/DEC or JOG in nature require that the command be also noted as part of a JOG type function key to perform properly. In addition, the system designer only provides those commands that are appropriate for a particular system. Care must be taken in preventing a command to be accessed by an operator which could cause machine damage, production loss, or human injury.

Applying these commands within the SC300 system requires absolute system familiarity. Please consult a qualified application/ design engineer if questions or conflicts arise in the use of these commands.

The other UC3 sub-command controls the setpoints of the drives. Again, setpoints are chosen by the system designer, and only those that are chosen can be manipulated by this sub-command. This command is also very dependent on the SDCS function key levels in DRIVES.IBM. They are, like UC3CM commands, not expected to be directly typed into the SC300 active console.

The first parameter of the UC3SET sub-command is a single character to inform SC300 of which setpoint to manipulate. When using this command, only the drive the operator is tabbed to can normally be manipulated. The characters allowed are :

1) 'W' - Selects the active regulator setpoint
2) 'X' - Setpoint is selected by the function key on level 7 of DRIVES.IBM
3) 'Y' - The speed setpoint of the peg drive is selected regardless of what value is in the tab register

The active regulator setpoint is the setpoint of the controlling regulator at the time 'W' is issued. The controlling regulator is designated by the last three letters of the MODE column for the drive selected by the TAB.

Level 7 of DRIVES.IBM allows the operator to select the setpoint to modify. In the case of 'X', it is crucial that this command is only executed from this level of function keys. The information of the actual regulator that is selected is stored within the function keys themselves.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

The peg drive is considered the "PACER" drive for the entire line. In other words, the speed of this drive determines the speed in which material travels through the line. In almost all cases, this drive is speed regulated only without the capability of modifying its speed through draw control or ratio control. It will never be switched into another regulator such as current or position. The 'Y' command is special because it selects the SPEED setpoint of the PEG drive as the setpoint to manipulate and does not require the TAB register.

The second parameter of the UC3SET sub-command is a single character that informs SC300 how to manipulate the setpoint. In general, the setpoint will be in some engineering units stored in the SC300 database. The characters allowed here are:

1) 'E' - Allows the operator to enter a number which replaces the present setpoint.
2) 'I' - Increments the setpoint by 1.0
3) 'D' - Decrements the setpoint by 1.0

By using 'E', the operator will be shown the present setpoint that he has selected. Then he is asked to enter a new number. Once he enters a new number he may accept it or reject it. If accepted, the old setpoint is replaced by this new number and is sent to the drive.

'I' will increment the setpoint by a whole number (1.0) without confirmation from the operator. This command is normally used in a function key defined as a JOG type. Similarly, 'D' will decrement the setpoint by a whole number, again without confirmation. The setpoint, once a change has been made, will be sent to the drive. All setpoints are limit checked, as is any value, as described by the UC6 commands. Any setpoint changed outside its limits revert to the original setpoint value and is not sent to the drives. The operator is given a message by SC300 that he has exceeded a limit and what those limits are.

Example:

The operator has selected function key level 4 (DRIVE OPERATE) in DRIVES.IBM. He has tabbed to a drive whose present controlling regulator is current. He selects a function key which has the command UC3SET WE behind it. (Note the space between UC3SET and WE; this space is required.) The operator has selected the current regulator of the drive he is tabbed on and now is prompted to enter a new current setpoint.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

There is a special setpoint change command that utilizes UC3CMZ instead of UC3SET. A single additional parameter is needed and that is either I or D. This command will increment or decrement (depending on I or D) the active regulator setpoint. This command must only be used in a direct communication mode to a single drive. Hence, this command should only be used on a function key level that only provides communication to one drive at a time (NOT LINE OR ZONE MODES, NOR PROGRESSIVE OR REGRESSIVE). SC300 will take the command, look at which regulator is active (the most outer loop) on the tabbed drive and reconfigure the command to set the UC3CMx y command that will increment or decrement that drive's regulator setpoint INTERNALLY TO THAT DRIVE.

Example:

The operator tabs to a drive that has as its active regulator: PROCESS. The operator issues UC3CMZ I by pressing the function key with that command assigned to it under menu level 4 (DRIVE OPERATE in DRIVES.IBM. SC300 intercepts the command and changes it into UC3CM2 4 (increment process regulator setpoint). The drive will then begin to increment the process regulator setpoint at the INC/DEC RATE assigned in EEROM or by a recipe (if configured as part of a recipe). When the operator releases the key, SC300 will automatically send UC3CM2 D (stop process regulator setpoint increment).

There are more commands accepted by SC300 to manipulate setpoints; however, these setpoints are accessed through the UC6 and UC19 commands which are explained elsewhere in this manual.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: UC4

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:
This command is used to re-calculate the value of the "Tab" register and send the new value to the Series Six.

Description:

This command is normally not required by user applications and is available for system development.

See Linc Fujita for more detailed description.

Example(s):

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: UC9

Sub-Commands:

UC90 : Select lead motor (you must be tabbed to the desired lead motor before you invoke this command)
UC91 : Select follow motor (you must be tabbed to the desired follow motor before you invoke this command)
UC92 : Blank draw display for this drive (tabbed drive)
UC93 : Show the draw for this drive as linear draw
UC94 : Show the draw for this drive as percent draw
UC95 : Make draw setups permanent (write changes to disk)
UC96 : Blank speed display for this drive
UC97 : Show the speed display for this drive

Parameters: None

Security Level: None

Short Description:

One of the columns of the Super Drive Control Screen (SDCS) is used to display the draw between two motors. (Draw is the difference in speed between any two drives in the same coordinated line.) These commands allows the operator to select which two motors are the basis of the draw display and display control of both draw values as well as speed feedback values. If a drive is tabbed, it will show the draw as defined by these commands. If a drive is not tabbed it will show its draw relative to the line peg motor.

Description:

These commands allow an operator to control the speed and draw displays on the SDCS screen for any and all drives displayed. These commands do not require invocation strictly from function keys although it is highly advisable to use them as part of the predefined level 13 of DRIVES.IBM. The DRAW column of the SDCS screen can show the difference of speed between any two motors. The tabbed drive is assumed as the drive the operator is interested in. The drive in which the tabbed drive's speed difference is taken from is the drive configured as its follow motor. Each drive's follow motor is defaulted to the line peg motor prior to any configurations done with these commands. The tabbed drive's motor is assumed to be the lead motor. When a drive is not tabbed, the draw column always will reflect the speed difference of that motor to the speed of the line peg.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Sub-Command: UC90

Although the tabbed drive is always considered the lead motor, invoking this command alerts SC300 that the operator wants to change speed or draw display configuration. The operator is prompted if the tabbed motor is really what he wants for the lead motor. If he answers 'Y', SC300 will remember that a lead motor has been selected. As long as the system is not reset, SC300 will not allow the operator to select another lead motor until a follow motor has been selected. The lead motor selection now allows the operator to tab to the follow motor (can be on another screen display but must be on an SDCS display).

Sub-Command: UC91

Once a lead motor has been selected, the operator tabs to the follow motor desired and then invokes this command. The operator will not be allowed to select a follow motor unless a lead motor has been selected first. Again, SC300 will ask if the follow motor selected is correct. If answered 'Y' the TAB will automatically switch back to the lead motor and the draw column display for that motor will show the speed difference between the tabbed motor and its newly selected follow motor. The system then becomes free for the operator to select another lead motor.

Sub-Command: UC92

If a motor has no tachometer or IS the line peg, the operator may not want to even see any draw value displayed for that motor. (If a drive has no tachometer, its CEMF value may be scaled and substituted as speed feedback.) If no display is desired, the operator simply issues this command and the draw column for that drive (he must TAB to it first) will go blank.

Sub-Command: UC93

There are two ways of showing speed difference in the draw column for any particular drive: Linear Draw and Percent Draw. These may be intermixed on any screen. This command configures the draw display to show Linear Draw or simply the speed of the lead motor minus the speed of the follow motor in the units of speed feedback of the lead motor.

Linear Draw = Lead Motor Speed Fdbk - Follow Motor Speed Fdbk

When this command is invoked, the value calculated by the algorithm above is immediately displayed regardless if the draw display was previously blanked.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Sub-Command: UC94

The second way of displaying draw is by showing the percentage difference between the lead motor speed and the follow motor speed in units of percent carried out to two decimal places. If the lead motor's speed should ever fall bellow the numerical value of 1.0, the draw column will show 0.00. The algorithm for percent draw is:

$$\text{Percent Draw} = \frac{\text{Lead Motor Speed FB} - \text{Follow Motor Speed FB}}{\text{Lead Motor Speed FB}} \times 100$$

Like UC93, when this command is invoked, the value calculated by the algorithm above is immediately displayed regardless if the draw display was previously blanked.

Sub-Command: UC95

All setups used with UC9 only updates the temporary memory. While the system is active, any changes made since starting SC300 will remain active. Once the system is stopped and re-started, all draw and speed display configurations are restored from a disk file (SC300.SMD). When this command is invoked, ANY and ALL changes made with UC9 commands will be saved to that disk file, permanently saving the changes until re-saved again.

Sub-Command: UC96

In some cases, (i.e. drive has no tachometer) it is not desired to show the speed feedback to the operator of a motor. This command will blank out the SPEED display column for the drive that is currently tabbed.

Sub-Command: UC97

This command restores a blank speed display to show the speed feedback of the tabbed drive. It may be used as a diagnostic tool if the speed feedback was actually CEMF and not normally shown. The CEMF feedback may temporarily be shown by invoking this command.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Notes on Draw Display

On the SDCS screen, when a drive is tabbed and the draw column shows a value (draw not blanked), there will be a yellow arrow pointing up or down indicating that this is the lead drive, and the follow drive, if the arrow is pointing up, is somewhere above it on this screen or on a previous screen. If the arrow is pointing down, the follow motor is somewhere below it on this screen or on a following screen display. The draw value that has a white arrow pointing up or down is the follow motor for the drive that is currently tabbed. The direction of the arrow for the follow motor is pointing to the relative position of the lead motor on this screen or another screen.

Remember that if a motor is not tabbed, the draw column is showing the draw for that motor relative to the line peg motor. You should never select a motor with its speed column blank as either the lead or the follow motor. Remember that draw values are not calculated in coordination with the lead and follow motors speed. This fact makes the draw value actually lag the displayed speeds shown for the lead and follow motors in some cases. Diagnostic evaluations should not be made from data in the draw column while drives are undergoing large excursions of speed. The use of trending the SPEED FEEDBACKS under the same time base should be used instead.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

IX. Recorder Command Group

The following commands are used to control the action of the background historical logging program. Historical logging can be selected on a point by point basis. The background recorder program can monitor up to 64 pens. The commands in this group allow the user to connect points to pens, disconnect pens, and save/recall various test setups to disk files.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: PEN

Sub-Commands:

a. PEN0 - Display screen 90 and current pen status
b. PEN1 - Return to previous display from display 90
c. PEN2 - Load setup from disk file
d. PEN3 - Save setup to disk file
e. PEN4 - Enable/Disable pen (same as PEN with no sub-commands)
f. PEN - Interactive enable/disable from any display

Parameters:

None

Security Level: 3

Short Description:

These commands control the logging of point data to disk files.

Description:

The PEN command without any subcommands will cause the system to immediately prompt the user for a tagname and check to see that it is defined in the TAGID34.IBM file (as loaded into memory). If the tagname is currently being recorded (connected to a pen), the user will be asked if the pen should be disconnected. If the tagname is not being recorded, a prompt asking for the desired pen number will be displayed. It is not possible to select a pen that is already in use. The pen sample rate is also requested and should be a number in seconds from 1 to 999.

The PEN0 command is used to automatically display screen 90 with the current status of all recording pens. This command MUST be used to display screen 90 or else no pen data will be shown.

The PEN1 command is normally used after completing other PEN commands. This command will return the screen to the screen in use prior to the first PEN command. This command should never be used unless display 90 is currently active.

The PEN2 and PEN3 commands allow the user to load/save all of the pen settings as shown on display 90. Files created using this command will have a default filetype of .PEN. When a new .PEN file is loaded, any existing pen settings are closed and the new ones are established.

Example(s):

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

?PEN0
This command will automatically display screen 90 with the currently active pen recorder settings.

?PEN

This command will provide a series of interactive prompts to enable or disable a pen channel.

?PEN1

This command will restore the previous display following a PEN0 command.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

X. Miscellaneous Command Group

The following commands are useful for operating the SC300 system but do not fall into any single category listed earlier. These commands allow running other programs, shutting down the system, user login/logout, window switching, and other similar tasks.

Most of these commands have no security associated with them. The exceptions are the QUIT and SUBMIT commands.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: QUI

Sub-Commands:

None

Parameters:

None

Security Level: 6

Short Description:

This command exits the SC300 console program and optionally shuts down all background programs.

Description:

When it is necessary to obtain the use of another virtual console for running application programs, it may be necessary to shut down the SC300 program. The QUI command shuts down the SC300 foreground task. As an option, it asks if ALL background tasks are to be shutdown as well.

If the answer is no, then data acquisition, pen recording, alarming, and remote recipe control are still possible.

If the answer is yes, the system is removed from the on-line state and will no longer communicate with the field devices.

Example(s):

?QUI

Requests that the SC300 console program shut down.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: HEL

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

This command will automatically call up display number 1.

Description:

This command is the system HELP command. The system has reserved display number 1 as the Help screen. This command will perform exactly the same function as the DIS1 command.

Example(s):

?HEL

Displays display number 1.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: SWI [a]

Sub-Commands:

None

Parameters:

a. Desired console number from 1 to 4

Security Level: None

Short Description:

This command will switch the active console to the requested console.

Description:

This command allows the user to switch the top level window (console) to another console. It is normally used in a function key program to allow automatic switching to the alarm console from the SC300 console. A SWI 1 command has no effect as the top level window is already active.

Example(s):

?SWI 3

Switches the top level console to window 3 to allow viewing the alarm summary data.

?SWI 2

Switches the top level console to window 2 to allow viewing the system analysis menu system.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: LOG

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

This command is used to login to the system and acquire pre-defined security priviledges.

Description:

The LOG command prompts for a username of up to 6 characters. The system then searches the SECURITY.IBM datafile for the available priviledges for this username. If two priviledge levels are supported, a second question is asked to determine whether the higher priviledge level is desired.

The system then sets the priviledge levels for function keys, display points, and typed commands.

If an auto-logoff timer feature has been enabled, the priviledge levels will be reset to 1 after the timer expires. The timer is reset after each successful command execution. Thus, the system will only logoff after a period of inactivity as defined in SC300.CFG.

The General Electric supplied SETPAS utility program is used to configure the database of usernames and their priviledges.

Example(s):

?LOG

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: LOC

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

This command logs the currently active user out and resets the priviledge levels back to 1.

Description:

This command logs the currently active user out and resets the priviledge levels back to 1. This command can be used to force an earlier logoff when the user does not wish to wait for the auto-logoff timer to expire.

Example(s):

?LOC

Logs the current user off the system and resets security levels to 1.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: SUB

Sub-Commands:

None

Parameters:

None

Security Level: 9

Short Description:

This command allows submitting a file of commands to the SC300 console for execution.

Description:

This command is automatically invoked at system startup to read a file of commands contained in the file SC300.DAT. Normally, these commands should never be re-executed.

However, it may be desirable to test some feature of the system using the SUB capability. If another virtual console is used to rename a test file of commands to SC300.DAT, then the SUB command will execute them.

Example(s):

?SUB

Automatically executes the list of commands found in the file named SC300.DAT.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: PRT [a]

Sub-Commands:

None

Parameters:

a. Device Characteristic

Security Level: None

Short Description:

This command prints the contents of the screen to the configured output device.

Description:

This command is used to print the current display to the printer number specified in the SC300.CFG configuration file. The default printer mode is also specified in this file.

The printer mode can be overidden using an optional device characteristic code from the list below:

K Print screen in color
L Print screen in small characters
T Print only the text on the screen (no graphics)

In addition, it is also possible to redirect the output of the print command to a file when the SC300.CFG file has been configured to do so.

Example(s):

?PRT

Prints the current display to the default printer device or file using the default printer characteristic. Defaults are specified in the SC300.CFG text file.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: FIR

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

This command allow the user to run another program while on-line using the same virtual console as SC300.

Description:

This command prompts for a standard Concurrent DOS command line and submits the command to the operating system command line interpreter for execution. The command is executed and then control of the console is returned to SC-300. Virtual console 1 is used to execute the command.

The acquisition of data, recording of data, alarms, etc., are unaffected by the use of this command (CPU time excluded).

Only commands that are .CMD, .COM, or .EXE can be requested to run using the FIR command.

Example(s):

?FIR

Asks for the program to run. User types SDIR *.DOC. The SDIR program is executed and control returns to SC300.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: FOR [a]

Sub-Commands:

None

Parameters:

a. Window number from 1 to 4

Security Level: None

Short Description:

This command allow the user to run another program while on-line using the same or another virtual console.

Description:

This command prompts for a standard Concurrent DOS command line and sends the command character string to the keyboard input buffer of the requested virtual console. The command is executed and then control of the console is returned to SC-300.

This command differs from the FIR command in that the command can be executed on any of the available virtual consoles. This capability allows for continued monitoring of the process while also performing the requested command in another window. In addition, the command string is not sent to the command line interpreter of the operating system. Instead, it is sent to the keyboard input buffer for the console. This means that an already running program may receive these characters as keystrokes.

The acquisition of data, recording of data, alarms, etc., are unaffected by the use of this command (CPU time excluded).

Example(s):

?FOR

Asks for the program to run. User types SDIR *.DOC. The SDIR program is executed and control returns to SC300.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: UTI

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

This command allow various utility functions to be performed. These include printer control, running programs, and showing the product version number.

Description:

The UTI command displays a menu of commands available. Each of these commands is invoked by typing the capitalized letter associated with each. Commands include:

Attach to the default printer device
Detach from the currently attached printer device
Set the default printer to a new device
Show the current default printer assignment
Show the current program version number and serial number
Run a task (identical to FIR command)

The printer control commands are usefull when more than one printer is available in the system. By using these commands, it is possible to redirect printer output without shutting down and restarting the SC-300 software.

Example(s):

?UTI

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88

Command: MEM

Sub-Commands:

None

Parameters:

None

Security Level: None

Short Description:

This command displays the available memory for storing displays and format data.

Description:

This command can be used to display the amount of memory remaining for use in buffering display and format data. If it is desired to buffer more or less data, the ADDHEAP utility program can be used with the SC300.CMD program.

The MEM command should be used during system development after displaying a screen with a large amount of data points to ensure that enough memory exists to buffer one display and one format.

Example(s):

?MEM

Shows the amount of available memory and the largest contiguous chunk in the memory space.

Copyright (C) 1988 General Electric Company, Drives Department, Erie, PA 16531 9/07/88